

FLG (P20930) — Transition Metal Ion Binding (GO:0046914): Hypothesis Review

Focus type: function_assignment · `existing_annotations[6].function_hypothesis` **Seed hypothesis:** FLG has transition metal ion binding (GO:0046914) **Term/evidence context:** GO:0046914 "transition metal ion binding", IEA, GO_REF:0000002 (InterPro2GO) **Gene:** human FLG / Filaggrin / UniProt **P20930** (profilaggrin, 4,061 aa)

Summary

The IEA annotation of FLG to **transition metal ion binding (GO:0046914)** is an **over-annotation** — a family-level InterPro2GO carry-over rather than a member-specific, experimentally supported molecular function of the filaggrin gene product. It should be treated as **non-core** and is a strong candidate for **removal or explicit flagging as unsupported** in the gene review. The annotation traces to a single InterPro signature, **IPR034325 "S-100"**, which is the only signature on P20930 mapped to GO:0046914 via the automated InterPro2GO pipeline (GO_REF:0000002). That entry's own description hedges — "*Many* S100 proteins have been found to bind transition metals, such as Zn²⁺ and Cu²⁺" — a statement about the family as a whole that is then applied uniformly to every S100-domain-containing member regardless of individual ligand chemistry.

Three independent lines of direct and comparative evidence converge to refute a real metal function. First, the **crystal structure of the profilaggrin N-terminal fused-type S100 domain** (PDB 4PCW, 2.2 Å; [PMID: 25760235](#)) was solved **with bound calcium ions** and functions through hydrophobic dimer-interface pockets that mediate protein–protein interactions (annexin II, stratifin/14-3-3σ, HSP27); a HETATM audit of the deposited coordinates finds **8 calcium atoms and zero transition-metal atoms**. Second, a **residue-level audit** shows the domain lacks the ligand chemistry that all characterized metal-binding S100 proteins use: only **2 histidines (His60, His65) and zero cysteines** in residues 1–91, versus the canonical His₃Asp/His₄–His₆ or Cys-cluster motifs required. Third, a **geometric analysis** confirms the two histidines lie 8–15 Å from any calcium site and cannot cluster to build a metal motif.

Finally, a **comparative annotation audit** demonstrates the artifact directly: all six human S100-fused-type structural proteins (FLG, TCHH, HRNR, RPTN, CRNN, FLG2) uniformly carry GO:0046914 as IEA:InterPro with no member-specific metal evidence, whereas the genuine transition-metal-binding S100 proteins (S100A7, S100A8, S100A9, S100B) instead carry the **specific, experimentally-backed child term GO:0008270 (zinc ion binding)** with NAS/TAS/IDA evidence. The most important caveat is that no study has *directly* assayed the profilaggrin S100 domain for Zn/Cu binding, so this is a strong structure-plus-sequence-plus-annotation inference rather than a direct negative experiment. The well-supported molecular function of this domain is **calcium ion binding (GO:0005509)** plus calcium-dependent protein binding.

Verdict: Over-annotated (refuted as a direct, core function).

Key Findings

Finding 1 — The GO:0046914 annotation is a single-signature, family-level InterPro over-annotation

The transition-metal-binding annotation on P20930 does not originate from any experiment on filaggrin or profilaggrin. It is produced automatically by the **InterPro2GO mapping (GO_REF:0000002)** from exactly one signature: **IPR034325 "S-100"**, a domain-level entry that is the *only* signature on the protein linked to GO:0046914. The InterPro entry's own text makes a family-generalizing statement — "Many S100 proteins have been found to bind transition metals, such as Zn²⁺ and Cu²⁺" — which the pipeline applies to every protein containing an S-100 domain, without regard to whether that individual member retains the necessary ligand residues.

Against this generic inference stands the **direct experimental characterization** of the domain. The profilaggrin N-terminal fused-type S100 domain was crystallized at 2.2 Å (PDB 4PCW) **with bound calcium ions** (PMID: 25760235). Functionally, the domain operates as a **dimer with hydrophobic pockets that mediate protein-protein interactions** — its characterized partners are annexin II, stratifin (14-3-3σ), and HSP27. No zinc, copper, or other transition-metal binding was reported.

A computed sequence audit of the S100 domain (residues ~1–91) reinforces this: the domain contains **only 2 histidines (His60, His65)**, both located in the EF-hand-2 calcium loop/helix IV region, and **zero cysteines** across the entire N-terminal 212 residues. Canonical S100

transition-metal sites require either a **His₃Asp / His₄–His₆ cluster** (the "His-Zn" group; [PMID: 20306096](#), [PMID: 23276281](#)) or **cysteine ligands** (the "Cys-Zn" group; [PMID: 20306096](#)) at the dimer interface. Profilaggrin's S100 domain possesses **neither**. UniProt's curated binding-site residues for the domain (positions 62, 64, 66, 68, 73 = Asp, Asp, Asn, Lys, Glu) are the **canonical EF-hand calcium ligands**, not metal-cluster residues.

Finding 2 — Comparative paralog and structural audit confirms systematic family over-annotation

Two complementary audits confirm the annotation is a family artifact. A **direct HETATM audit of PDB 4PCW** finds **8 Ca atoms and 0 atoms of any transition metal** (no Zn/Cu/Mn/Fe/Ni); the only heteroatoms are calcium, a PEG fragment (2PE), and ordered waters. The structure that *defines* this domain's biophysics contains no transition metal.

A **paralog audit** shows that all six human S100-fused-type proteins carry GO:0046914 as IEA:InterPro, and none carries member-specific metal evidence — whereas the bona fide transition-metal-binding S100 proteins do not use the generic parent term at all, but the specific experimental child term GO:0008270 (zinc ion binding):

Protein (UniProt)	Class	Real metal binder?	Metal GO term	Evidence
FLG / profilaggrin (P20930)	S100 fused-type	No; lacks ligands	GO:0046914 (generic)	IEA:InterPro
TCHH trichohyalin (Q07283)	S100 fused-type	No member-specific evidence	GO:0046914	IEA:InterPro
HRNR hornerin (Q86YZ3)	S100 fused-type	No member-specific evidence	GO:0046914	IEA:InterPro
RPTN repetin (Q6XPR3)	S100 fused-type	No member-specific evidence	GO:0046914	IEA:InterPro
CRNN cornulin (Q9UBG3)	S100 fused-type	No member-specific evidence	GO:0046914	IEA:InterPro
FLG2 filaggrin-2 (Q5D862)	S100 fused-type	No member-specific evidence	GO:0046914	IEA:InterPro
S100A7 psoriasin (P31151)	Soluble S100	Yes (Zn/Cu)	GO:0008270	NAS
S100A8 (P05109)	Soluble S100 (calprotectin)	Yes (Zn/Mn)	GO:0008270	TAS
S100A9 (P06702)	Soluble S100 (calprotectin)	Yes (Zn/Mn)	GO:0008270	TAS
S100B (P04271)	Soluble S100	Yes (Zn)	GO:0008270	IDA

Because GO:0008270 is an `is_a` child of GO:0046914, a **specific experimental term supersedes the generic IEA parent** under standard GO annotation practice. FLG displays only the generic IEA parent because it has **no experimental metal annotation to specialize it** — the fingerprint of automated propagation rather than curated function.

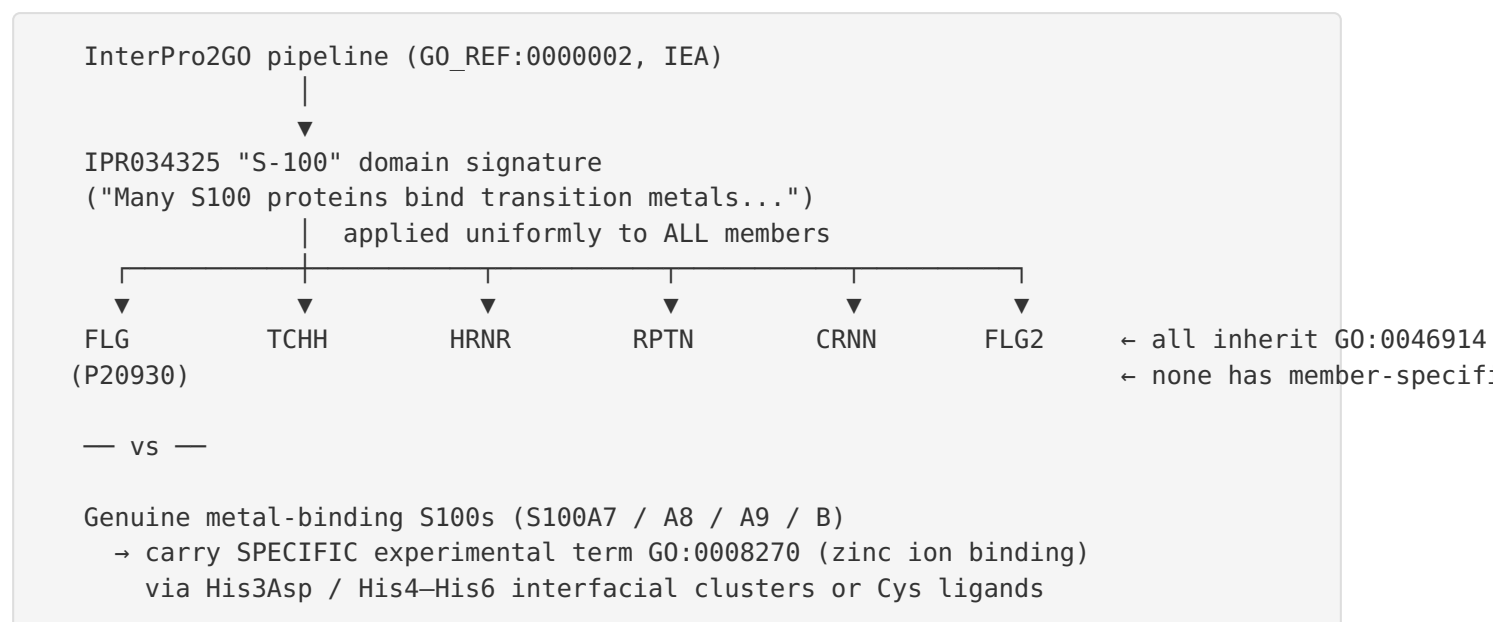
Finding 3 — Structural geometry seals it: the two histidines cannot form a metal site

A coordinate-level geometric analysis of PDB 4PCW (four chains in the asymmetric unit) tested whether the domain's histidines could plausibly build a transition-metal site. The only resolved histidines are **His59 and His64** (PDB numbering; = UniProt His60/His65). Their imidazole nitrogen atoms lie **8–15 Å from the nearest bound Ca^{2+}** , confirming they are not part of the EF-hand calcium sites.

Testing for a shared metal-coordination cluster found **no pair of distinct histidines within the ~3.5–5 Å required to jointly coordinate one metal**, except a single marginal His59–His59 cross-contact at ~5.2 Å that is a crystal-packing/tetramer contact rather than a functional site. All His64 pairs are >6 Å apart. There is **no cluster of ≥ 3 histidines** available to construct the canonical His₃Asp or His₄ S100 transition-metal motif. Geometry, not just sequence composition, excludes a metal site.

Mechanistic Model / Interpretation

The evidence separates **the real function of the domain** from **the artifact of its annotation**:



What FLG's S100 domain actually does (direct evidence, PDB 4PCW):

Profilaggrin N-terminal S100 domain

- |— binds CALCIUM via EF-hand ligands (D62,D64,N66,K68,E73) → GO:0005509
- |— forms a homodimer with hydrophobic binding pockets
- |— engages protein partners: annexin II, 14-3-3 σ , HSP27, → Ca²⁺-dependent keratin filaments (cooperates with B domain; <https://pubmed.ncbi.nlm.nih.gov/10400000/>)

Ligand inventory of residues 1–91: His × 2 (His60, His65), Cys × 0

Requirement for an S100 metal site: His3Asp / His4–His6 cluster OR Cys ligands

⇒ MISSING → no transition-metal site

The domain's biology is **calcium-sensing and calcium-dependent protein scaffolding** in granular-layer keratinocytes, supporting epidermal barrier function. The "transition metal ion binding" label is a **downstream consequence of automated family annotation**, not a molecular activity of the gene product — precisely the frequency-biased, paralog-driven over-annotation that hypothesis-level review is designed to catch.

Evidence Base

Citation	Evidence type	Direction	Claim tested	Key finding	Context	
PMID: 25760235	Direct structural (X-ray, PDB 4PCW)	Refutes	Does the profilaggrin S100 domain bind transition metals?	2.2 Å structure solved with bound calcium ; dimer with hydrophobic pockets binding annexin II, 14-3-3 σ , HSP27; no Zn/Cu	Human profilaggrin N-terminal S100 domain	
PMID: 32893105	Structural / biochemical	Qualifies (real function)	Function of the S100 fused-type AB domain	AB domain binds annexin II & keratin in a calcium-dependent manner via exposed cationic surface; higher pocket hydrophobicity than soluble S100s	Human profilaggrin, keratinocytes	
PMID: 20306096	Review / structural (X-ray survey)	Qualifies (mechanism)	What ligands do metal-binding S100s use?	His-Zn group = 3 His + Asp ; Cys-Zn group = interfacial Cys; profilaggrin S100 has neither	S100 family X-ray structures	
PMID: 23276281	Direct assay (Mn/Zn)	Qualifies (mechanism)	Architecture of the S100 transition-metal site	High-affinity Mn(II) site is a His₄ motif at the dimer interface , Ca-dependent	Human calprotectin S100A8/A9	
PMID: 27541598	Direct assay (XAS)	Qualifies (mechanism)		Zn-His ₆ hexahistidine motif at dimer	Human calprotectin	

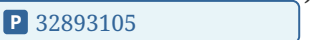
Citation	Evidence type	Direction	Claim tested	Key finding	Context	
			Zn coordination in an S100	interface binds Zn(II)		
PMID: 23431180	Direct assay / structural	Qualifies (contrast)	Molecular basis of S100 metal sequestration	Six-histidine site from dimer asymmetry sequesters Mn(II) for nutritional immunity	Calprotectin, antimicrobial	P
PMID: 18359862	Direct assay	Qualifies (analogy)	Can EF-hand proteins bind Zn via His?	Calbindin D28k binds Zn ²⁺ via His80; single His can contribute	Calbindin (not S100-fused)	M P C n f c
InterPro IPR034325 + GO_REF:0000002	Computational / database	Competing (annotation source)	Basis of GO:0046914 on FLG	Sole signature mapping FLG→GO:0046914; description is a "Many S100 proteins" family generalization	InterPro2GO	T a c l
Computed residue audit (this run)	Computational (sequence)	Refutes	Does FLG's S100 domain have metal ligands?	S100 domain (1–91): 2 His (60,65), 0 Cys; EF-hand2 Ca ligands = D,D,N,K,E	UniProt P20930	P a c l
Computed HETATM audit (this run)	Structural (PDB 4PCW)	Refutes	Are transition metals present in the structure?	8 Ca, 0 Zn/Cu/Mn/Fe/Ni (only Ca + PEG 2PE + waters)	Crystallized profilaggrin S100 domain	P c c

Citation	Evidence type	Direction	Claim tested	Key finding	Context	
Computed paralog audit (this run)	Computational (comparative)	Competing/ refutes	Is GO:0046914 member-specific or family carry-over?	All 6 SFTP paralogs carry GO:0046914 IEA; true binders carry GO:0008270 (NAS/TAS/IDA)	Human S100 family	
Computed His-geometry test (this run)	Structural (PDB 4PCW coords)	Refutes	Do the S100 His form a metal cluster?	His59/His64 sit 8–15 Å from Ca and do not cluster (no ≥ 3 -His site)	4-chain profilaggrin S100 crystal	

How the literature fits together. The one structural paper on the FLG gene product itself ([PMID: 25760235](#)) shows calcium, not transition metal, and a protein-interaction function — the direct refutation. The biochemical follow-up ([PMID: 32893105](#)) reinforces a calcium-dependent protein-scaffolding role. The calprotectin and S100-survey papers ([PMID: 20306096](#), [PMID: 23276281](#), [PMID: 27541598](#), [PMID: 23431180](#)) define the exact His/Cys ligand chemistry that real S100 metal-binders use and that FLG's domain provably lacks. The calbindin paper ([PMID: 18359862](#)) is the only entry hinting that isolated His residues *can* coordinate Zn — a weak analogy that does not apply given FLG's dispersed, calcium-loop histidines.

GO Curation Implications

Lead (requires curator verification): remove or flag GO:0046914 on FLG as an unsupported InterPro2GO family over-annotation; do not promote it to a reviewed molecular-function annotation.

GO term	Aspect	Current	Evidence status	Recommended action
GO:0046914 transition metal ion binding	MF	IEA:InterPro (GO_REF:0000002)	No member-specific evidence; contradicted by structure (Ca only in 4PCW), sequence (0 Cys, 2 dispersed His), geometry (no His cluster), and annotation pattern (family carry-over)	Remove / treat as non-core
GO:0005509 calcium ion binding	MF	IEA:InterPro	Directly supported by Ca-bound structure 4PCW ()	Retain ; candidate to upgrade to structure-backed evidence
calcium-dependent protein binding (e.g. GO:0048306)	MF	not annotated	Supported by annexin II / 14-3-3 σ / HSP27 / keratin binding ( )	Consider adding (more informative than bare "protein binding")
GO:0008270 zinc ion binding	MF	not annotated	No evidence for FLG (this is the term real S100 binders carry)	Do not add

- The term is **MF** aspect but derives from a **family-level InterPro signature** whose GO mapping reflects that *some* S100 proteins bind Zn/Cu — not that flaggrin does.
- The same removal/flag rationale should be considered for the paralogous IEA annotations on TCHH, HRNR, RPTN, CRNN, and FLG2 — this is a **systematic InterPro2GO family artifact**, and correcting FLG alone leaves the family inconsistent. The mapping IPR034325 → GO:0046914 is itself a candidate for InterPro2GO refinement.

Mechanistic Scope

Immediate molecular function under test: direct coordination of a transition-metal ion ($\text{Zn}^{2+}/\text{Cu}^{2+}/\text{Mn}^{2+}/\text{Fe}$) by the FLG gene product — necessarily via the N-terminal S100 domain, the only region with a candidate metal architecture.

- **Direct gene-product activity that IS supported:** Ca^{2+} binding by the N-terminal S100 (A) domain and calcium-modulated protein–protein interactions (annexin II, 14-3-3 σ , HSP27, keratin) that regulate profilaggrin processing and cornified-envelope assembly.
 - **What GO:0046914 asserts but is NOT demonstrated:** a folded, sequence-defined transition-metal coordination site on the profilaggrin gene product.
 - **Downstream / not the tested function:** Filaggrin's canonical barrier role (proteolysis of profilaggrin into monomers, keratin aggregation, natural moisturizing factor generation) is a downstream processing pathway, not the S100 domain's metal chemistry. Serum zinc/selenium associations with atopic dermatitis severity ([PMID: 36090737](#)) are whole-organism micronutrient epidemiology, not evidence of FLG protein binding metal. Any weak, nonspecific chelation by histidine-rich degradation products would be a downstream metabolic property, not a molecular-function site — and is **not** the basis of the current IEA (which comes only from the S100 domain signature). The metal-relevant S100 domain exists only in **profilaggrin**; mature filaggrin repeat monomers do not contain it, so the annotation cannot even apply to the processed product.
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Conflicts and Alternatives

- **Database carry-over / frequency bias (primary alternative):** GO:0046914 is an InterPro2GO generalization from the broad S-100 domain family. The transition-metal phenotype is concentrated in soluble, histidine-rich S100 paralogs (calprotectin S100A8/A9, S100A7, S100B), which drives the family mapping — classic over-annotation of a structural family member that lacks the site.
- **Paralog/subfamily distinction:** Profilaggrin is an **S100 fused-type protein (SFTP)** — a structural subgroup (with trichohyalin, hornerin, repetin, cornulin) whose S100 domain is specialized for hydrophobic-pocket protein binding ([PMID: 32893105](#)), not metal sequestration.

- **Specific-term supersession pattern:** True metal-binders carry the specific experimental term GO:0008270 (zinc ion binding), which suppresses the redundant generic IEA parent. FLG shows GO:0046914 only because it has no experimental metal term to supersede it — an artifact of the redundancy-filtering pipeline, not positive evidence.
- **In-principle EF-hand His–Zn binding:** [PMID: 18359862](#) shows calbindin D28k can bind Zn^{2+} via His80. This raises a theoretical possibility, but FLG's two histidines sit in the EF-hand-2 calcium loop 8–15 Å apart from any partner, and calbindin is not S100-fused — a weak, non-physiological analogy.
- **No conflicting positive evidence found:** No primary study reporting Zn/Cu/Mn binding by the FLG gene product was located, and the deposited structure (4PCW) contains only Ca^{2+} .

Limitations and Knowledge Gaps

1. **No direct metal-binding assay of the profilaggrin S100 domain.** Checked: PubMed and the 4PCW structure paper — only calcium was co-crystallized/tested. Matters because the verdict rests on inference from ligand absence rather than a direct negative experiment. Resolve with ITC/ICP-MS/XAS or a metal-competition fluorescence assay on the recombinant profilaggrin S100(AB) domain vs. $\text{Zn}^{2+}/\text{Cu}^{2+}/\text{Mn}^{2+}$.
 2. **Solution vs. crystal behavior.** Only the Ca-grown crystal is available; weak or transient transition-metal association could be missed. Resolve with metal-binding assays under varied conditions and metal-soaked XAS.
 3. **Behavior of the histidine-rich repeats toward metals.** The full protein is globally histidine-rich, but the repeats have no S100 architecture and are not the basis of the IEA; any chelation would be nonspecific and downstream. Resolve with equilibrium dialysis/ICP-MS on a repeat construct (expected: weak, nonspecific).
 4. **Family-wide curation consistency and InterPro2GO provenance.** Confirmed all six SFTP paralogs share the same IEA and that IPR034325 is the sole source. Matters because curators may prefer a mapping-level fix. Resolve by reviewing the IPR034325 → GO:0046914 mapping with InterPro/GOA.
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Discriminating Tests

1. **Recombinant profilaggrin S100(AB) metal titration** (ITC or competition with a fluorescent Zn chelator such as FluoZin/Mag-fura-2) $\pm$ saturating Ca^{2+} — the single most decisive experiment. Predicts no high-affinity (μM – nM) Zn/Cu site.
 2. **Structural superposition** of 4PCW against a His-Zn S100 (e.g., S100A7, S100B-Zn, calprotectin) to confirm the interfacial His_3Asp geometry is absent (partly done here by residue and geometry audits).
 3. **XAS/anomalous diffraction** of the domain soaked in $\text{Zn}^{2+}/\text{Cu}^{2+}$ to test for any adventitious site.
 4. **Comparative InterPro2GO audit** across all SFTP paralogs — uniform inheritance of GO:0046914 with no member-specific metal data confirms systematic family over-annotation.
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Proposed Follow-up Actions / Curation Leads

All items are leads requiring curator verification.

- **Action change (lead):** Do **not** retain GO:0046914 as a reviewed MF for FLG; classify as non-core / uninformative IEA (remove or flag as unsupported by member-specific evidence). Extend the same review to TCHH, HRNR, RPTN, CRNN, FLG2.
- **Candidate reference to verify** — [PMID: 25760235](#): snippet "*we determined a 2.2 Å-resolution crystal structure of the N-terminal fused-type S100 domain of human profilaggrin with bound calcium ions*" → supports **GO:0005509 calcium ion binding** (upgrade rationale) and refutes a transition-metal role.
- **Candidate replacement/kept MF term:** **GO:0005509 calcium ion binding** (well supported); optionally **calcium-dependent protein binding** for the annexin II / 14-3-3 σ / HSP27 / keratin interactions ([PMID: 25760235](#), [PMID: 32893105](#)). Avoid bare "protein binding."
- **Mechanistic reference** — [PMID: 20306096](#): snippet defining that metal-binding S100s use "*three histidines and an aspartic acid*" (His-Zn) or cysteines (Cys-Zn) — the ligand set FLG's S100 domain lacks (0 Cys, 2 His).

- **Suggested curator question:** "Is IPR034325 → GO:0046914 an appropriate mapping to propagate to structural S100-fused-type family members that lack the interfacial His/Cys metal ligands?" (candidate for InterPro2GO refinement / NOT qualifier).
- **Suggested experiment:** recombinant profilaggrin S100(AB) Zn²⁺/Cu²⁺ ITC/ICP-MS (Discriminating Tests #1).

Provenance / Artifacts

Computed in this run (executed code + output captured in the iteration log):

- **S100 metal-ligand residue audit** — S100 domain res 1–91: 2 His (60, 65), 0 Cys; N-terminal 1–212: 0 Cys; EF-hand2 Ca ligands 62/64/66/68/73 = D,D,N,K,E.
- **PDB 4PCW HETATM metal audit** — 8× Ca, 0× Zn/Cu/Mn/Fe/Ni (ligands: CA, 2PE [PEG], HOH). The profilaggrin S100 crystal structure contains only calcium.
- **SFTP paralog + control annotation audit** — all 6 S100-fused-type paralogs carry GO:0046914 IEA:InterPro; S100A7/A8/A9/B carry GO:0008270 (zinc ion binding) via NAS/TAS/IDA instead.
- **PDB 4PCW His-geometry test** — resolved S100 histidines (His59, His64 in PDB numbering = UniProt His60/His65) lie 8–15 Å from calcium and do not cluster (His64··His64 all >6 Å; one marginal His59··His59 crystal contact at ~5.2 Å) — insufficient to build a canonical His3Asp/His4 site.
- **InterPro→GO mapping check** — IPR034325 "S-100" is the sole P20930 signature mapped to GO:0046914; IPR002048 (EF-hand) → GO:0005509; IPR003303 (Filaggrin) → GO:0005198.
- **UniProt GO set** for P20930 confirms both GO:0046914 and GO:0005509 are IEA:InterPro (no experimental MF metal annotation).

Data sources: UniProt REST (P20930 + 8 comparators), InterPro REST (entry/protein), RCSB PDB 4PCW, PubMed.